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FLITS: an interactive, reproducible workbench for fast radio burst analysis

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Software

- Review
- Repository
- Archive

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Summary

Fast radio bursts (FRBs) are brief flashes of radio emission, usually detected from distant galaxies (Lorimer et al., 2007; Petroff et al., 2022). They are commonly studied using a dynamic spectrum: a record of the radio signal as a function of time and observing frequency. Measurements of burst brightness, duration, frequency structure, and polarization help researchers investigate the source and the material through which its radiation has travelled.

FLITS (Fast-Look Interactive Transient Suite) is an open-source Python package with a browser interface for analysing detected bursts. It brings data inspection, interference removal, measurement, and modelling into a single interactive session. It reads the data, masks interference, and fits the `fitburst` (Fonseca et al., 2024) burst model itself, so users can apply each step to the same selected data without moving between separate programs. Other analysis packages can work on the same prepared data, which is available from Python and in exported files.

FLITS saves the measurement settings, selected data regions, excluded frequency channels, and calibration inputs in a portable session file. Together with the original data, this file allows users to reopen an analysis and recompute supported measurements. Outputs describe how uncertainties were estimated and flag missing calibration information, making published results easier to check and reproduce.

Statement of need

For each detected burst, radio observations provide dynamic spectra, often for the four Stokes parameters: total intensity (I), linear polarization (Q and U), and circular polarization (V). Some observations retain only total intensity. Turning these data into an astrophysical interpretation requires measurements such as dispersion measure (DM), which describes the delay of lower-frequency radiation by free electrons, and rotation measure (RM), which describes the wavelength-dependent rotation of linear polarization by magnetized plasma. Other quantities include burst width, arrival time, spectrum, fluence (flux density integrated over time), and, given calibration and distance information, peak luminosity. Scintillation bandwidth and scattering time characterize fine frequency structure and pulse broadening caused by propagation through intervening material.

These derived properties are routinely reported in papers, but a table of values rarely records every choice needed to reproduce them. Results depend on the burst and background regions, excluded interference, time and frequency resolution, calibration, and fitted model. Comparing

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observations from instruments with different formats and sensitivities adds further complexity. The campaign that motivated FLITS combines data from the Green Bank Telescope at L and P band, Nançay, Westerbork, Onsala, Stockert, and CHIME.

Automated pipelines are essential for samples of hundreds to thousands of bursts, but unusual or particularly informative events often warrant individual attention. For example, successive components in some repeating FRBs appear at progressively lower frequencies, the "sad-trombone" effect (Hessels et al., 2019). This structure can be confused with an incorrect dispersion correction, so the DM that maximizes signal-to-noise need not best preserve the burst's components. Interactive inspection lets researchers compare DM choices, separate overlapping components, select uncontaminated background data, and check what a fitted model fails to explain. It supports careful assessment of such ambiguities without claiming to remove them automatically.

FLITS combines this flexibility with a record of the settings used. Users can refine an individual analysis, compare measurements across observations, and share the saved session and original data so collaborators can examine how the results were obtained.

State of the field

Several packages address the analysis of bursts after detection. `fitburst` (Fonseca et al., 2024) models dynamic spectra to estimate burst shape, dispersion, and scattering parameters. FRBGui (Chamma et al., 2023) provides interactive measurements of burst structure in time and frequency. DM_phase (Seymour et al., 2019) estimates DM using resolved burst structure, while RM-Tools (Van Eck et al., 2026) provides rotation-measure synthesis and polarization fitting. Stingray (Bachetti et al., 2024) offers broader time-series and variability analysis, including power spectra, correlations, and statistical modelling of astronomical light curves. These tools provide complementary ways to study transient signals.

FLITS brings data preparation and its own implementations of several of these methods together in a shared session. Its readers are tested against `your` (Aggarwal et al., 2020), its automatic masking follows the per-channel statistical tests of `jess` (Kania et al., 2026), and its burst-model fits are tested against `fitburst`. Its own RM-synthesis implementation is checked against an RM-Tools (Van Eck et al., 2026) reference dataset. A separate workbench is useful because these methods are otherwise spread across several specialist packages and file formats, each preparing the data separately. New file formats are added as reader plugins without changes to FLITS itself, and other tools can work on the session's prepared data through the Python interface or exported files. Its main contribution is to keep an analysis's selections and settings alongside its measurements, so the interactive workflow can be inspected and supported calculations repeated.

Software design

The same Python analysis routines serve both the browser interface and command-line workflows. Changes made in the browser update a session containing the data and analysis settings; the calculations can also run without opening a browser. This allows automated tests to exercise the measurement routines directly.

For larger samples, `flits measure` applies the same steps to a list of files without a browser: it masks interference, locates the burst in time and frequency, and measures signal-to-noise, peak flux density, fluence, widths, and arrival times. Each result records a status and warning flags, so bursts whose automatic selections need checking can be reopened in the interface. `flits demo` downloads a reference burst and opens it in the browser.

FLITS reads SIGPROC filterbank files and PSRFITS search and folded data (Hotan et al., 2004), and supports CHIME/FRB HDF5 products, including public catalogue dynamic spectra

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(CHIME/FRB Collaboration, 2021) and beamformed power data. These readers convert the input into common arrays and descriptive information. Additional file formats can be supported through reader plugins.

A saved JSON session file records the input-file location and SHA-256 checksum, DM, data crop, burst and background regions, frequency range, excluded channels, calibration inputs, and available results. This takes more care to maintain than a table of measurements, but preserves the settings needed to revisit an analysis. On reopening, FLITS checks the source data against the recorded checksum and metadata. The file captures the saved state, not a history of every interaction; reproducing the calculation also requires the original data and an appropriate software environment.

The `flits replay` command reopens the data and recomputes the core measurements and any recorded width comparisons, frequency-drift measurements, and polarization analyses performed on the session data. It can compare the core numerical results with their saved values within a specified tolerance. The saved drift-analysis settings include the Monte Carlo random seed. DM scans, other time- and frequency-structure analyses, and optional model fits remain saved results rather than being rerun by this command.

Exports use JSON for structured results, CSV for tables, NumPy NPZ archives for numeric and string arrays, PNG/SVG for plots, and optional SIGPROC files for selected data windows. They do not serialize Python objects with pickle. These formats allow stored results to be inspected independently of FLITS, but do not guarantee identical recalculation after software changes. Snapshots and exports automatically record FLITS, Python, and installed dependency versions, with source revisions where available. Each analysis retains its originating software record when a session is reopened or other measurements are recomputed; missing provenance in older files remains explicitly unknown. Tests use files generated by earlier releases to check preservation of stored results and detect changes during recalculation. Reproducing an analysis still requires preserving its software environment: version records identify software but do not archive it. The repository supplies pinned dependencies and a public reference workflow.

Trial DMs can be scored by signal-to-noise, by the burst-structure criterion of DM_phase (Seymour et al., 2019), or by the smoothed-structure metric of Sutinjo et al. (2023), which also reports the range of DMs that the noise cannot distinguish from the best one.

Polarization analysis accepts suitable full-Stokes polarization input files with known conventions, or separately prepared Q/U spectra. FLITS performs weighted RM synthesis (Brentjens & Bruyn, 2005; Heald, 2009), which combines the polarization across frequency to estimate Faraday rotation. Instrumental polarization calibration and corrections for Earth's ionosphere must be applied separately. Flux and fluence uncertainties distinguish measurement noise from calibration uncertainty. When uncertainty in the system-equivalent flux density (SEFD), a measure of telescope sensitivity, is not supplied, the outputs flag the incomplete uncertainty budget. These checks help users judge a result's limits; they do not establish its scientific validity on their own.

User documentation and reproducible examples are available at https://dirkkuiper.github.io/flits/.

Research impact statement

FLITS made the burst measurements for a study of the brightest bursts of the repeating FRB 20240114A observed with several telescopes (Kuiper et al., in prep.), and J. Huang uses it in a separate study of the same source (Huang et al., in prep.). It is also used in a pulsar study of subpulse morphology and DM optimization (Kuiper et al., in prep.) and by A. Moroianu to analyse how the DM and burst morphology of precisely localized repeating FRBs evolve over time (Moroianu et al., in prep.). C. Makarytchev and V. Shah use it to determine accurate DMs for bursts from FRB 20240209A.

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Researchers outside the development team have contributed code. M. A. Fine added support for folded PSRFITS pulsar data, treating pulse phase as the time axis so that pulse profiles can be modelled. J. Huang corrected how FLITS reads polarization data from the Nançay Radio Telescope.

The public GBT reference workflow provides a downloadable burst cutout, a saved session, expected measurements, and pinned dependencies. Automated checks exercise both the published package and the current source checkout against this reference.

AI usage disclosure

Generative AI tools (OpenAI Codex/ChatGPT, Anthropic Claude Code, Google Gemini, and Microsoft Copilot) assisted with software development, documentation, and manuscript preparation, including prose editing and review. Specific model versions were not systematically recorded; however, use generally involved a then-current flagship model from the respective provider. The authors reviewed and validated AI-assisted outputs, made the core scientific and design decisions, and retain responsibility for the work.

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FLITS builds on the scientific Python ecosystem, particularly NumPy (Harris et al., 2020), SciPy (Virtanen et al., 2020), Astropy (Astropy Collaboration, 2022), and Matplotlib (Hunter, 2007). Its readers and interference masking replace `your` (Aggarwal et al., 2020) and `jess` (Kania et al., 2026), which earlier versions used.

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